Extended reading — Week 7: Policy Gradients and REINFORCE

November 8, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Williams (1992): Simple Statistical Gradient-Following Algorithms

Primary source or book record: https://doi.org/10.1007/BF00992696.

What to learn. Likelihood-ratio estimators turn a sampled return into a stochastic policy-gradient direction.

What the result requires. Interchanging differentiation and expectation needs regularity and integrability. A finite sampled gradient need not improve return, and an unbiased direction is not a global convergence proof.

Notebook investigation. Use `reinforce_loss`. Check the sign: a positive advantage should increase log probability when the optimiser minimises the negative surrogate.

2. Sutton et al. (1999): Policy Gradient Methods with Function Approximation

Primary source or book record: https://proceedings.neurips.cc/paper/1999/hash/464d828b85b0bed98e80ade0a5c43b0f-Abstract.html.

What to learn. Relates performance gradients to policy derivatives and action values under an appropriate state weighting. Compatible approximation supplies a particular relationship between critic fitting and the gradient.

What the result requires. An arbitrary neural critic is not automatically compatible. State the discounted or average-reward objective and its occupancy convention; dropping time discount factors can change the estimator.

Notebook investigation. Derive the score identity from normalisation of the action distribution. Then identify the state weighting needed for the objective used in the local episodic loop.

3. Greensmith, Bartlett and Baxter (2004): Variance Reduction Techniques

Primary source or book record: https://jmlr.org/papers/v5/greensmith04a.html.

What to learn. Analyses baselines and variance in gradient estimation. The variance-minimising baseline weights returns by squared score magnitude.

What the result requires. A state value baseline need not equal the variance-minimising baseline, and a poor baseline can increase variance. Baseline independence from the sampled action is central to the simple zero-mean argument.

Notebook investigation. Use `optimal_baseline`. Compute $b^* = \sum_i w_i G_i / \sum_i w_i$ for weights $(1, 4)$ and returns $(2, 5)$: 4.4, not the ordinary mean 3.5.

4. Chen et al. (2019): Top-K Off-Policy Correction

Primary source or book record: https://arxiv.org/abs/1812.02353.

What to learn. The inclusion probability under K independent draws with replacement followed by deduplication is $1 - (1 - p)^K$. Differentiating gives the factor $K(1 - p)^{K-1}$.

What the result requires. This is not sampling exactly K distinct items without replacement; deduplication can yield fewer items. The expression depends on the sampling mechanism, not specifically on softmax, and is not an exact propensity for a constrained slate.

Notebook investigation. Use `topk_offpolicy_multiplier`. At $p = 0.2, K = 3$, inclusion probability is 0.488 and its derivative with respect to p is 1.92. Check a finite difference.

5. Gao, Schulman and Hilton (2022): Scaling Laws for Reward Model Overoptimization

Primary source or book record: https://arxiv.org/abs/2210.10760.

What to learn. Studies optimisation of a proxy reward using a synthetic gold reward model as a reference. More proxy optimisation can worsen the reference objective.

What the result requires. The gold model is not actual human welfare. The fitted scaling relationships differ between reinforcement learning and best-of-N sampling; there is no universal quadratic law in KL or universal safe KL setting.

Notebook investigation. Sketch proxy and held-out reward against optimisation effort. Mark where their trends separate. Explain why a small training loss alone cannot diagnose reward alignment.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.

Statistics reading bridge

An unbiased policy-gradient sample can still be mostly noise. Use expectation and sample-mean variance to inspect complete score-weighted gradients. Reconstruct the baseline identity from the policy-gradient primary source, then test variance using independent repeated trajectories rather than a single favorable update.

Checked textbook locations: Siegrist § § 4.3, 4.7, 6.2, PDF pp.460, 507, 817. https://stats.libretexts.org/@go/page/10158. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.